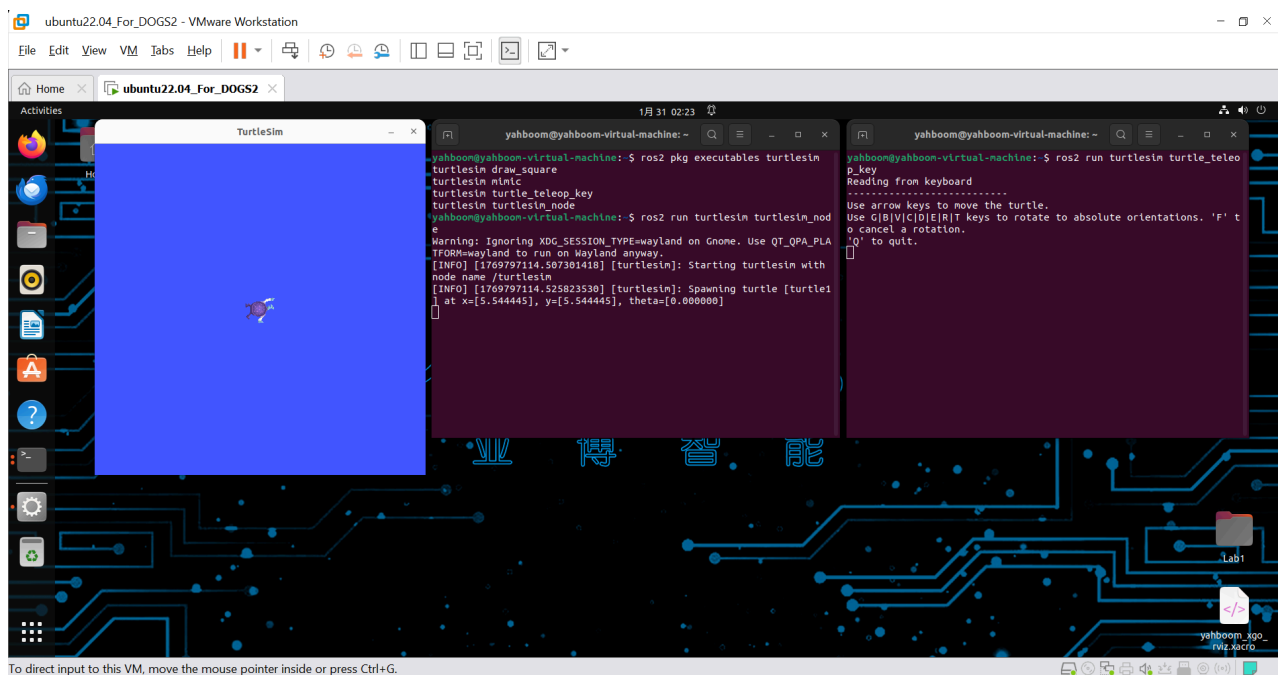


Lab 2 Individual Deliverables

Task 2:

Starting Nodes:

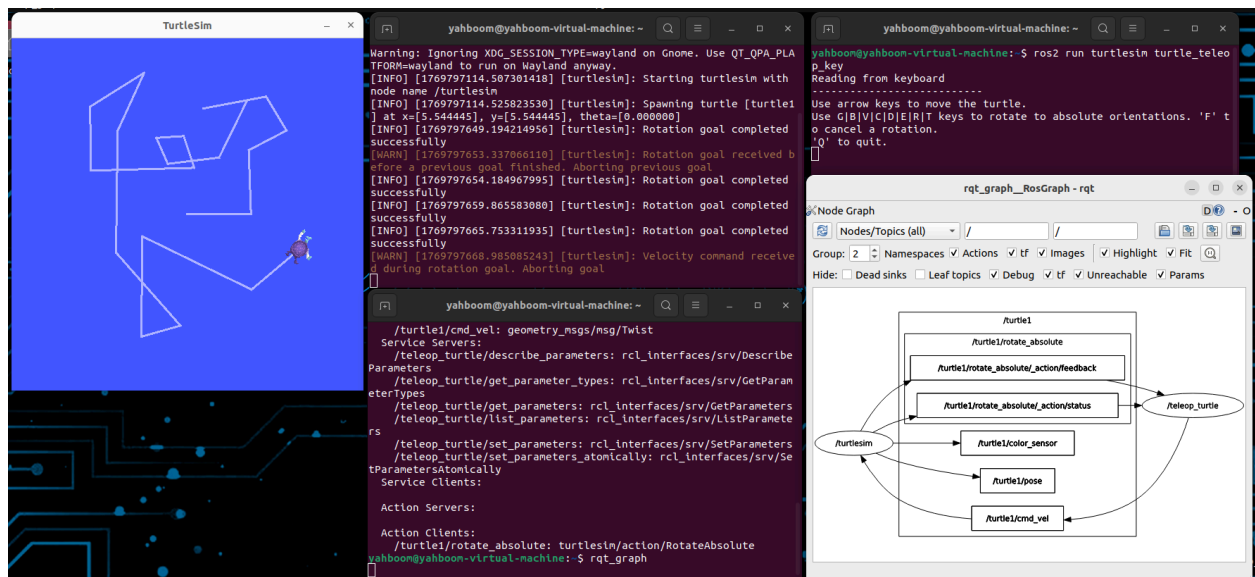
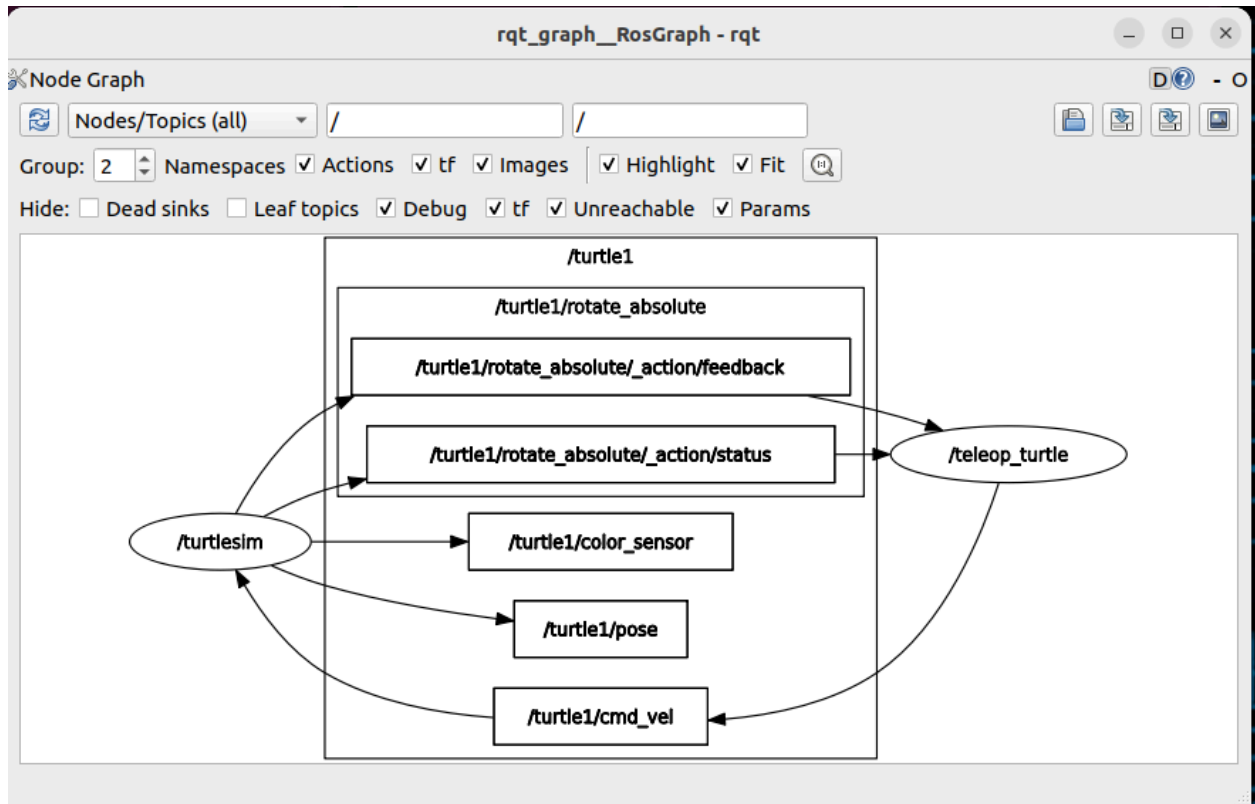
Screenshot showing the multiple terminals



Topics:

Screenshots for steps 7, 21, and 25

Step 7:

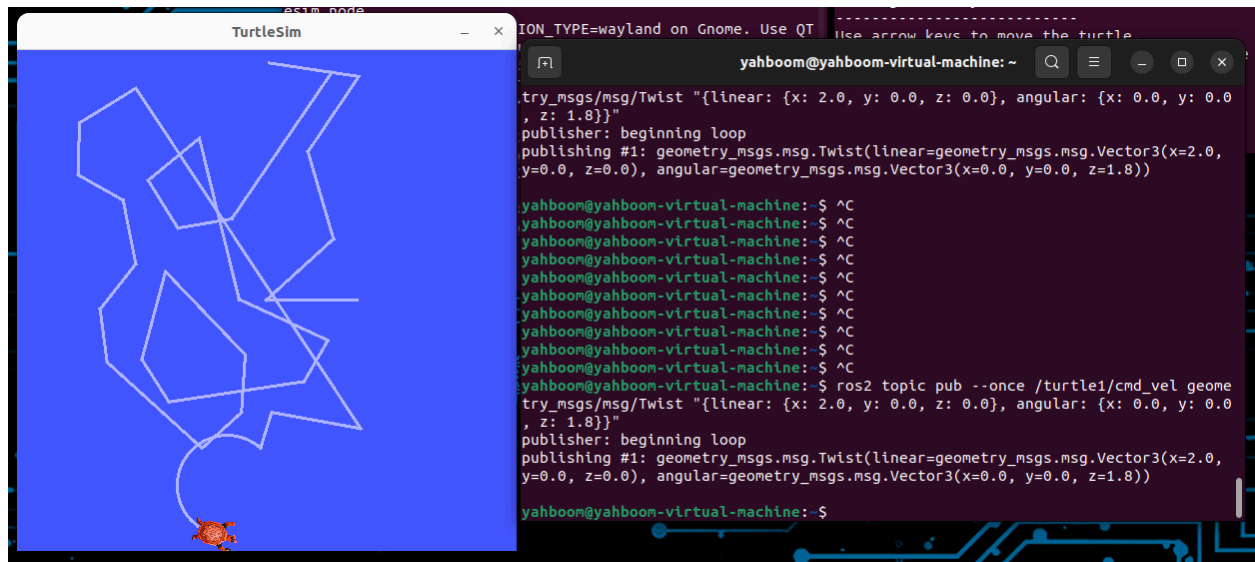


Step 21:

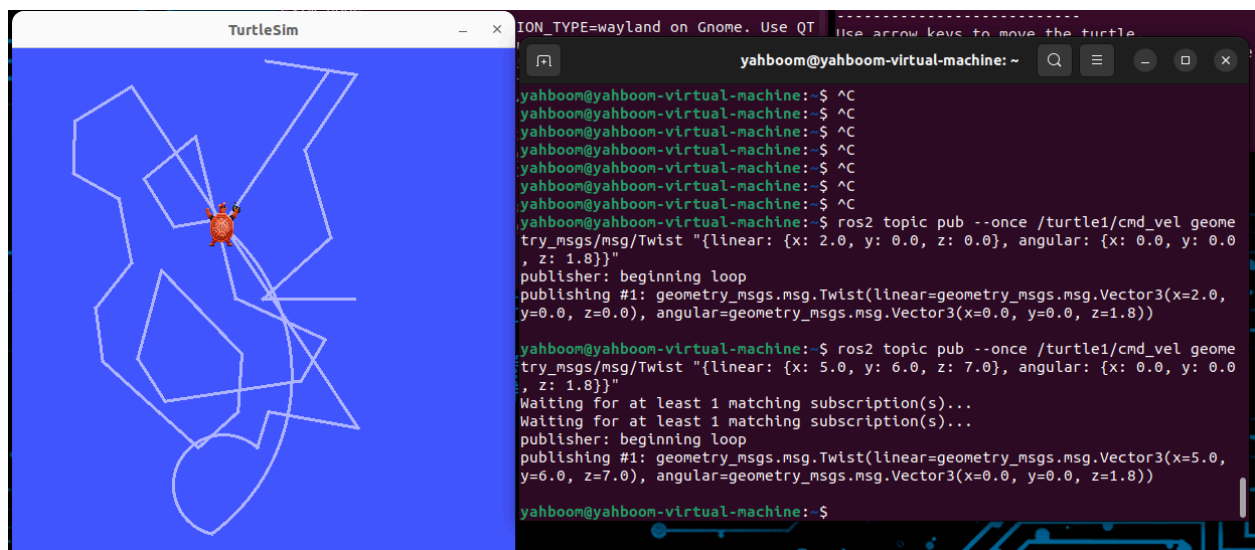
```
yahboom@yahboom-virtual-machine: ~  
---  
^Cyahboom@yahboom-virtual-machine:~$ ros2 topic info /turtle1/pose  
Type: turtlesim/msg/Pose  
Publisher count: 1  
Subscription count: 0  
yahboom@yahboom-virtual-machine:~$ ros2 interface show turtlesim/msg/Pose  
float32 x  
float32 y  
float32 theta  
  
float32 linear_velocity  
float32 angular_velocity  
yahboom@yahboom-virtual-machine:~$ ros2 topic echo /turtle1/pose  
x: 5.417484283447266  
y: 2.258474349975586  
theta: 2.3967411518096924  
linear_velocity: 0.0  
angular_velocity: 0.0  
---  
x: 5.417484283447266  
y: 2.258474349975586  
theta: 2.3967411518096924  
linear_velocity: 0.0  
angular_velocity: 0.0  
---  
x: 5.417484283447266  
y: 2.258474349975586  
theta: 2.3967411518096924  
linear_velocity: 0.0  
angular_velocity: 0.0  
---  
x: 5.417484283447266  
y: 2.258474349975586
```

Step 25:

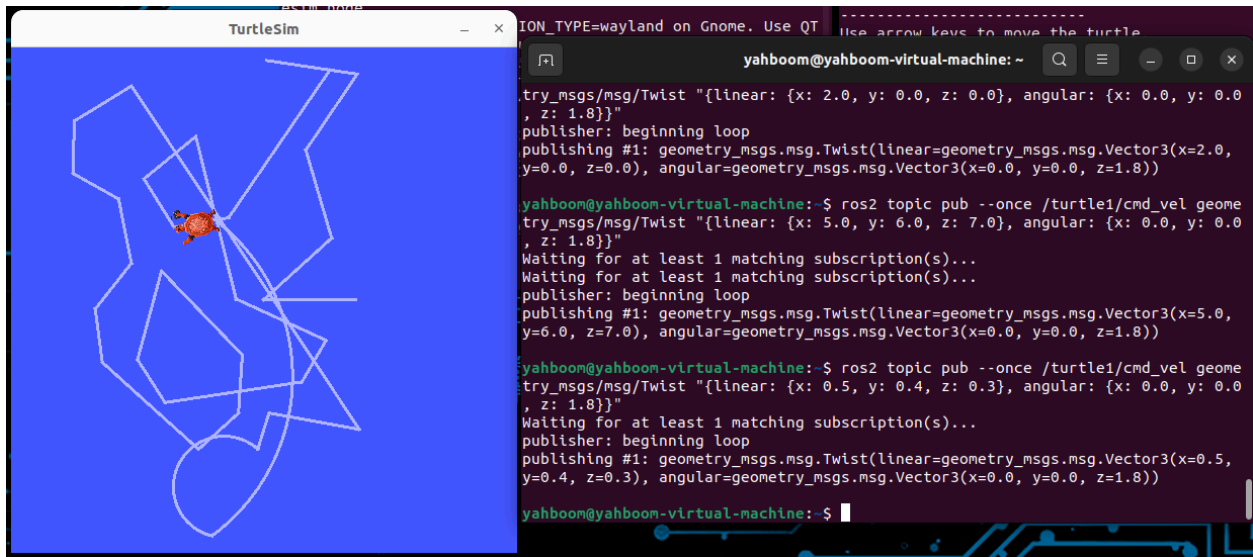
For reference (before running the unique tests):



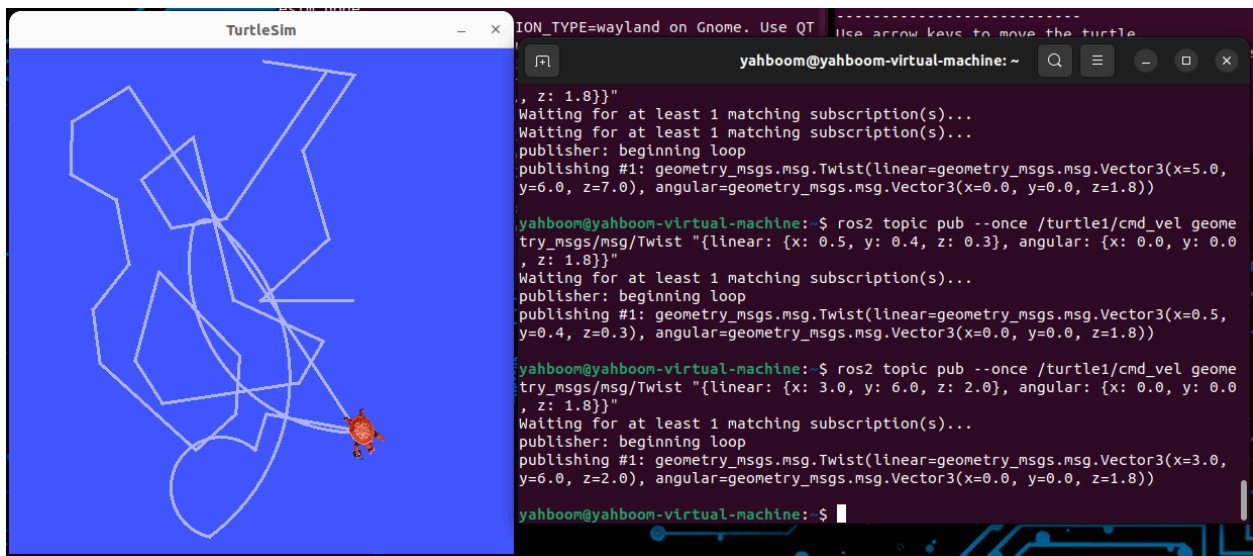
1. With X=5, Y=6, Z=7:



2. With $X=0.5$, $Y=0.4$, $Z=0.3$:

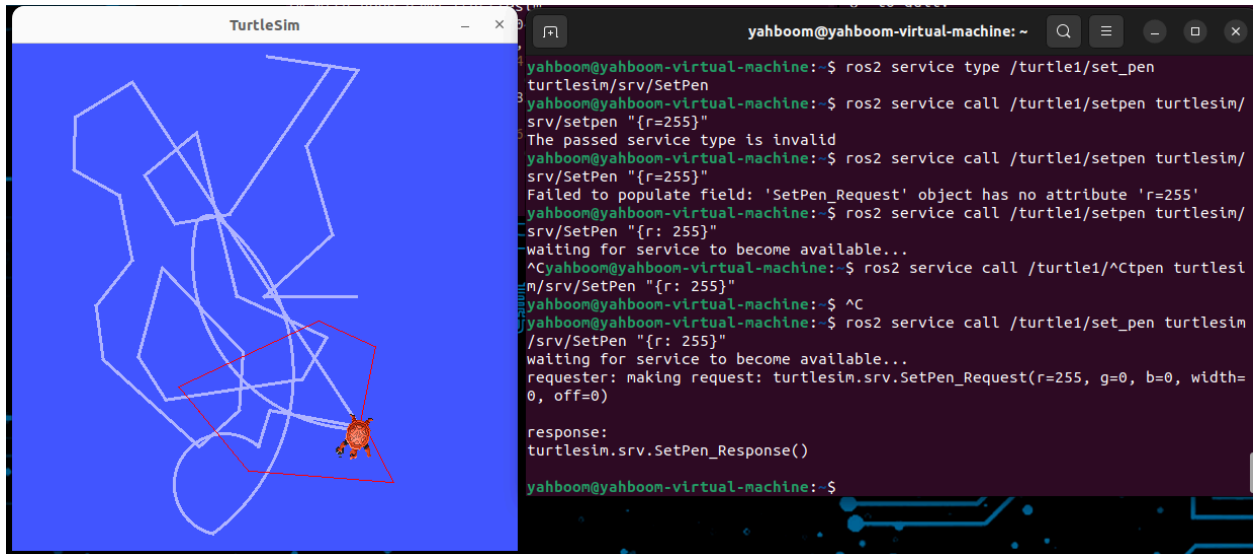


3. With $X=3$, $Y=6$, $Z=2$:



Services:

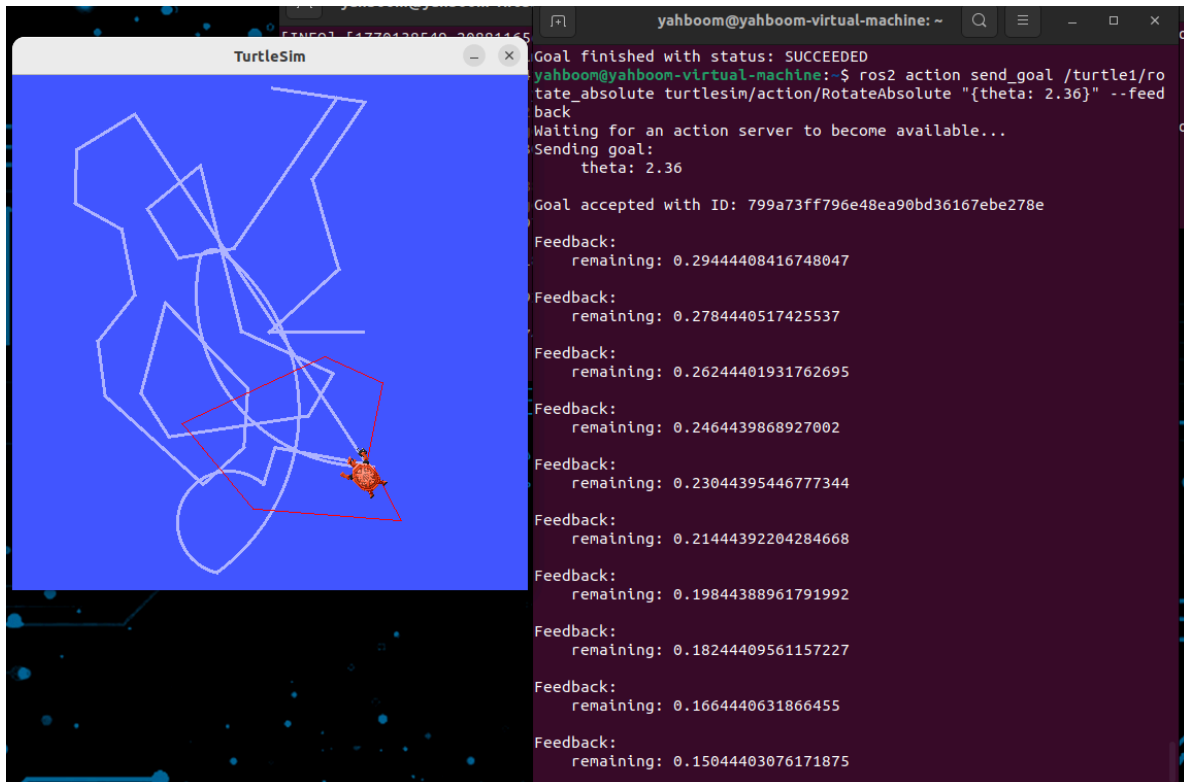
Screenshot of the Challenge

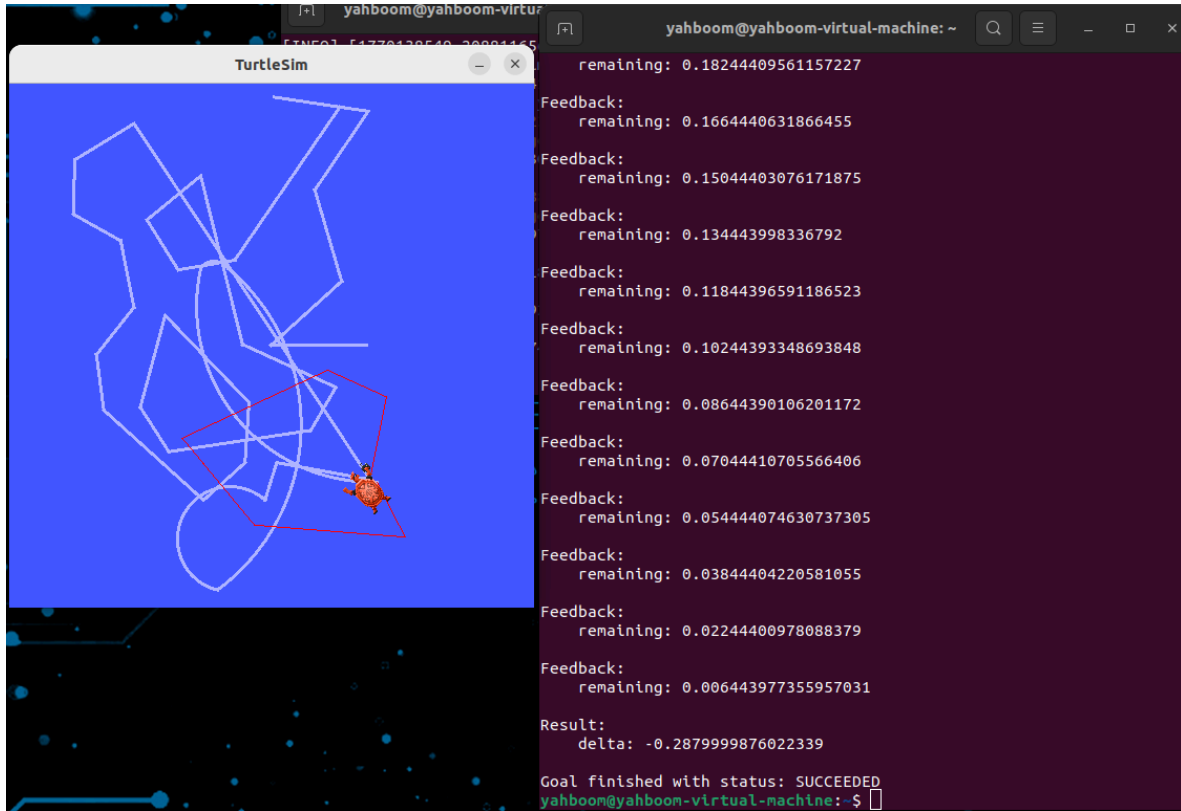


Action:

Executing step 8 with a unique goal position

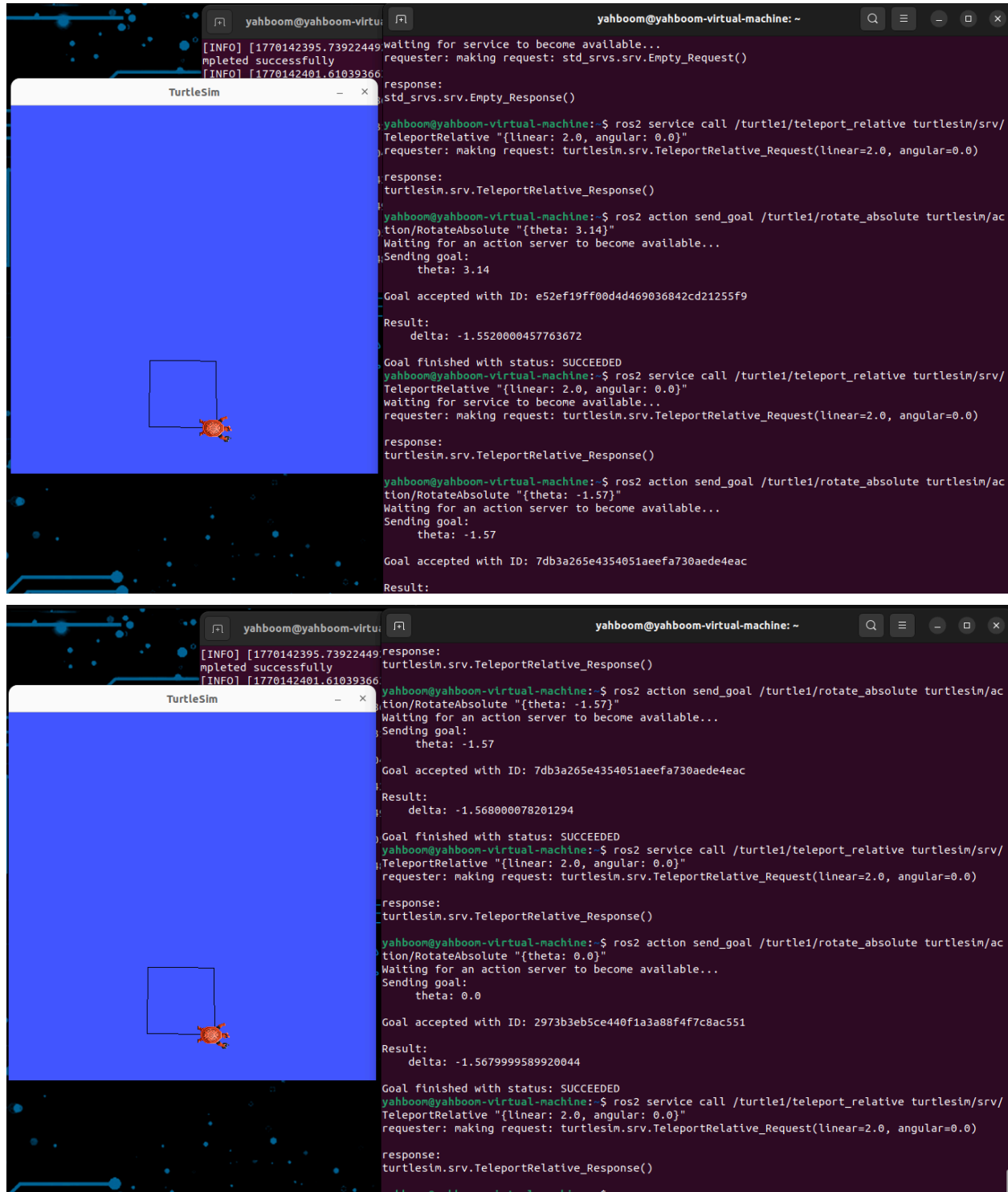
With theta = 2.36:





Task 3:

Final image and text with a brief explanation of the method used.



The image displays two screenshots of a ROS2 environment, showing a terminal window and a TurtleSim window. The terminal window is titled 'yahboom@yahboom-virtual-machine: ~' and the TurtleSim window is titled 'TurtleSim'.

First Screenshot:

```
[INFO] [1770142395.73922449] waiting for service to become available...
[INFO] [1770142401.61039366] requester: making request: std_srvs.srv.Empty_Request()
response:
std_srvs.srv.Empty_Response()
yahboom@yahboom-virtual-machine:~$ ros2 service call /turtle1/teleport_relative turtlesim/srv/
TeleportRelative "{linear: 2.0, angular: 0.0}"
requester: making request: turtlesim.srv.TeleportRelative_Request(linear=2.0, angular=0.0)
response:
turtlesim.srv.TeleportRelative_Response()
yahboom@yahboom-virtual-machine:~$ ros2 action send_goal /turtle1/rotate_absolute turtlesim/ac
tion/RotateAbsolute "{theta: 3.14}"
Waiting for an action server to become available...
Sending goal:
theta: 3.14
Goal accepted with ID: e52ef19ff08d4d469036842cd21255f9
Result:
delta: -1.5520000457763672
Goal finished with status: SUCCEEDED
yahboom@yahboom-virtual-machine:~$ ros2 service call /turtle1/teleport_relative turtlesim/srv/
TeleportRelative "{linear: 2.0, angular: 0.0}"
waiting for service to become available...
requester: making request: turtlesim.srv.TeleportRelative_Request(linear=2.0, angular=0.0)
response:
turtlesim.srv.TeleportRelative_Response()
yahboom@yahboom-virtual-machine:~$ ros2 action send_goal /turtle1/rotate_absolute turtlesim/ac
tion/RotateAbsolute "{theta: -1.57}"
Waiting for an action server to become available...
Sending goal:
theta: -1.57
Goal accepted with ID: 7db3a265e4354051aeefa730aede4eac
Result:
```

Second Screenshot:

```
[INFO] [1770142395.73922449] response:
turtlesim.srv.TeleportRelative_Response()
[INFO] [1770142401.61039366] yahboom@yahboom-virtual-machine:~$ ros2 action send_goal /turtle1/rotate_absolute turtlesim/ac
tion/RotateAbsolute "{theta: -1.57}"
Waiting for an action server to become available...
Sending goal:
theta: -1.57
Goal accepted with ID: 7db3a265e4354051aeefa730aede4eac
Result:
delta: -1.568000078201294
Goal finished with status: SUCCEEDED
yahboom@yahboom-virtual-machine:~$ ros2 service call /turtle1/teleport_relative turtlesim/srv/
TeleportRelative "{linear: 2.0, angular: 0.0}"
requester: making request: turtlesim.srv.TeleportRelative_Request(linear=2.0, angular=0.0)
response:
turtlesim.srv.TeleportRelative_Response()
yahboom@yahboom-virtual-machine:~$ ros2 action send_goal /turtle1/rotate_absolute turtlesim/ac
tion/RotateAbsolute "{theta: 0.0}"
Waiting for an action server to become available...
Sending goal:
theta: 0.0
Goal accepted with ID: 2973b3eb5ce440f1a3a88f4f7c8ac551
Result:
delta: -1.5679999589920044
Goal finished with status: SUCCEEDED
yahboom@yahboom-virtual-machine:~$ ros2 service call /turtle1/teleport_relative turtlesim/srv/
TeleportRelative "{linear: 2.0, angular: 0.0}"
requester: making request: turtlesim.srv.TeleportRelative_Request(linear=2.0, angular=0.0)
response:
turtlesim.srv.TeleportRelative_Response()
yahboom@yahboom-virtual-machine:~$
```

The TurtleSim window shows a blue square representing the environment. A small orange turtle icon is visible in the bottom right corner of the square. The background of the terminal window is dark purple with a blue and white pattern.

Brief Explanation:

I created a square by alternating between instantaneous translation and monitored rotation. I used the service "teleport_relative" to draw the straight line of a fixed length (2.0) in the direction that the turtle is facing. To rotate the turtle and create a "corner," I used the "ros2 action send_goal" command to trigger the "rotate_absolute" action, which turned the turtle to a specific orientation in radians. Using the action instead of a simple command or keyboard key provides a feedback loop, ensuring that the turtle accurately completed its rotation before initiating the command to create the next side of the square.